Candidate's name._____**Sign** ____

Personal number _____

UACE CHEM P 2 **SET 12**

EDUCATOR'S FORUM

Uganda Advanced Certificate of Education

Uganda Advanced Certificate of Education

CHEMISTRY PRACTICAL (PAPER 2)

Time: 3 hours 15 minutes

INSTRUCTIONS TO CANDIDATES:

- This examination paper consists of two compulsory examination items.
- Responses to these items must be written in the answer booklet provided.
- Begin each item on a fresh page.
- All working must be clearly shown. Graph papers will be provided.

For examiners use only:

COMPETENCE	P	M	Dp	Da	O	C&R
ITEM 1						
ITEM 2						

ITEM 1

A food aroma chemical manufacturing plant synthesizes flavor compounds from agricultural materials. A technician accidentally mislabeled a distillation flask containing a sweet-smelling, non-acidic liquid code-named K3. The plant manager suspects the compound is a ketone that could be utilized in manufacturing artificial pear essence, but they must confirm it is not an aldehyde or an ester before it can be mixed into the commercial batch. You are provided with 10 cm^3 of liquid K3.

Task: Design and carry out a systematic qualitative analysis investigation to identify the functional group family and carbon backbone saturation level of sample K3. Perform a minimum

of 6 distinct diagnostic tests with clear observations and logical deductions. Based on your findings, write its IUPAC name, structural formula, and explain its reactive properties in standard laboratory environments.

ITEM 2

A metallurgical workshop utilizes hydrochloric acid to remove rust and scale from iron sheets before galvanisation (pickling). The spent pickling liquor containing unreacted hydrochloric acid and iron(II) chloride is usually treated before disposal. The environmental compliance team needs to determine the exact concentration of the residual free hydrochloric acid in a batch of spent liquor to calculate the precise amount of lime required for neutralisation. The team requires a fast, standard volumetric analysis, but the intense green color of the iron(II) ions interferes with direct color-change indicators during standard direct titrations. A back-titration using excess standard sodium carbonate is proposed.

Provided Materials: You are provided with the following materials:

- **FA1:** Spent pickling liquor sample (containing residual HCl)
- **FA2:** 0.10 mol dm^{-3} standard nitric acid solution
- **Solid Z3:** 2.00 g of pure anhydrous sodium carbonate (Na2CO3)
- **Indicator:** Methyl orange indicator
- **Solvent:** Distilled water

Relevant reactions

- Reaction of the Spent Pickling Liquor (FA1) containing Free HCl with Excess Sodium Carbonate Solution (Solid Z_3) dissolved)

$$Na_2CO_3(aq) + 2HCl(aq) \rightarrow 2NaCl(aq) + CO_2(g) + H_2O(l)$$

- Back-Titration of Unreacted (Excess) Sodium Carbonate with Nitric Acid (FA2)

$$Na_2CO_3(aq) + 2HNO_3(aq) \rightarrow 2NaNO_3(aq) + CO_2(g) + H_2O(l)$$

Task

Design and carry out a quantitative back-titration experiment to determine the concentration (in mol dm^{-3}) of the unreacted free hydrochloric acid in the FA1 liquor sample. Present your titration data in a clear table showing at least six accurate titre values, and show all steps of your calculations, including equations for all intermediate and neutralization reactions involved.

END